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Noise-Robust DEMON Spectrum Extraction for Ship-Radiated Noise via Adaptive Decomposition and Sparse Reconstruction

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Abstract

To improve the quality of detection of envelope modulation on noise (DEMON) spectra extracted from ship-radiated noise under noisy conditions, a noise-robust DEMON spectrum extraction method incorporating adaptive decomposition and sparse reconstruction is proposed. Ensemble empirical mode decomposition (EEMD) is employed to adaptively select the effective frequency band, while the diagonal slice of the third-order cumulant is utilized to suppress Gaussian noise and enhance modulation characteristics. Subsequently, sparse Bayesian learning (SBL) is introduced to reconstruct the DEMON spectrum and improve the spectral resolution. The effectiveness of the proposed method is validated through numerical simulations and lake-trial experiments. The results show that the proposed method produces cleaner spectral backgrounds and improves the clarity of the shaft frequency and its harmonics compared with the conventional EMD-based method, especially under low modulation-depth conditions. These results demonstrate that the proposed method provides an effective solution for robust DEMON spectrum extraction from ship-radiated noise in complex underwater acoustic environments.

Keywords: ship-radiated noise; DEMON spectrum; ensemble empirical mode decomposition; third-order cumulant; sparse Bayesian learning

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1. Introduction

Passive acoustic target recognition is an important research topic in underwater acoustics. Among various feature extraction methods, the detection of envelope modulation on noise (DEMON) spectrum is widely used to extract modulation components generated by shaft rotation and propeller motion. These modulation features contain important information, such as shaft frequency and blade passing frequency, and have been extensively employed for underwater target classification and recognition [1–3]. However, the performance of DEMON analysis strongly depends on the quality of the extracted modulation signal and is susceptible to background noise interference, especially under low signal-to-noise ratio conditions.

In recent years, various DEMON-based approaches have been investigated to improve modulation feature representation and underwater acoustic signal analysis. Hashmi and Raza proposed a novel DEMON spectral analysis method combined with an empirical

knowledge-based reference criterion for acoustic signal classification [2]. Li and Wang further developed a feature self-supervised learning method driven by DEMON spectral models for underwater ship noise recognition [4]. Although these studies provide effective strategies for improving DEMON-based feature extraction, learning-based methods usually require sufficient training data, whereas traditional signal-processing methods still face challenges in extracting weak modulation components under severe noise interference.

To improve the quality of DEMON spectra, adaptive decomposition methods have attracted increasing attention in underwater acoustic signal processing. Empirical mode decomposition (EMD) was originally proposed for nonlinear and non-stationary signal analysis [5]. Subsequent studies revealed that EMD exhibits filter-bank-like characteristics and is capable of adaptively decomposing signals into intrinsic mode functions (IMFs) distributed in different frequency ranges [6]. These advantages make EMD suitable for underwater acoustic signal analysis and signal detection [7]. Applications of EMD have also been extended to ship-radiated noise analysis and target classification, demonstrating the potential of decomposition-based methods for acoustic feature extraction [8,9].

Despite its advantages in adaptive decomposition, EMD suffers from the mode mixing problem, which may lead to inaccurate separation of signal components. To alleviate this limitation, ensemble empirical mode decomposition (EEMD) was introduced by incorporating white-noise-assisted analysis, thereby improving decomposition robustness [10]. The improved decomposition capability of EEMD has promoted its application in underwater acoustic target detection and signal processing under complex acoustic environments [11]. Beyond EEMD, other decomposition strategies have also been explored. Complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) improves decomposition completeness through adaptive noise assistance, whereas variational mode decomposition (VMD) extracts band-limited components based on variational optimization [12,13]. These developments have expanded the applicability of decomposition techniques in underwater acoustic analysis. Recent studies have further demonstrated the effectiveness of EMD-based methods for characterizing nonlinear and time-varying underwater acoustic signals, while also highlighting the need to address issues such as mode mixing and noise sensitivity [14]. Despite their capability of adaptive component separation, decomposition-based methods usually do not directly incorporate noise suppression mechanisms, and residual noise may remain in the extracted components, thereby affecting subsequent DEMON spectrum extraction.

Higher-order statistics (HOS) have been extensively investigated for signal detection and parameter estimation [15,16]. Since higher-order cumulants are theoretically insensitive to zero-mean Gaussian noise, they have been widely employed for the analysis of non-Gaussian signals and weak signal detection [17,18]. In addition, bispectral analysis and higher-order spectral estimation provide effective tools for revealing nonlinear interactions and extracting useful information from noisy observations [19,20]. These properties make higher-order spectral techniques suitable for enhancing weak modulation features embedded in background noise.

In underwater acoustic applications, Gao et al. proposed a DEMON spectrum extraction method based on an improved EMD approach and 1.5-dimensional spectrum analysis, demonstrating the potential of combining adaptive decomposition and higher-order spectral techniques for modulation feature enhancement [21]. However, such methods mainly focus on exploiting the noise-resistant characteristics of higher-order spectra, while the subsequent refinement of DEMON spectral resolution is not considered.

Matrix decomposition methods, such as singular value decomposition (SVD), have been widely investigated for signal detection and feature extraction in noisy environments [22]. Sparse Bayesian learning (SBL) has attracted considerable attention because of its excellent sparse representation capability and high-resolution reconstruction performance [23]. Theoretical developments of SBL have further established its advantages in basis selection and sparse approximation problems [24–26]. Moreover, SBL has been extended to more complicated sparse recovery scenarios and has demonstrated considerable potential for high-resolution spectral estimation and weak signal reconstruction [27,28]. Owing to its sparse representation capability, SBL provides an effective approach for estimating closely spaced spectral components and improving spectral resolution. Most existing SBL-based frameworks are developed for sparse recovery problems with predefined dictionaries or signal models, and their performance may be affected when prior information is inaccurate or signal components are severely contaminated by noise. The combination of SBL with noise-robust modulation feature enhancement provides a potential way to further improve DEMON spectrum representation.

Although existing studies have improved DEMON spectrum extraction from different perspectives, adaptive decomposition, noise-resistant feature enhancement, and spectral resolution improvement are usually investigated separately. Decomposition methods may suffer from residual noise, higher-order spectral methods mainly focus on noise suppression, and sparse reconstruction methods usually depend on predefined models. This indicates that an integrated framework that combines these advantages for robust DEMON spectrum extraction is still required.

In this paper, an integrated DEMON spectrum extraction framework is developed by combining EEMD, HOS analysis, and SBL. The developed framework aims to improve modulation feature extraction and spectral representation under complex acoustic environments. The remainder of this paper is organized as follows. Section 2 introduces the ship-radiated noise model and the corresponding modulation characteristics. Section 3 presents the theoretical framework of the proposed method. Numerical simulations are conducted in Section 4, and the application of the proposed method to measured data is illustrated in Section 5. Finally, the conclusions are summarized in Section 6.

2. Signal Model

In this section, the characteristics of ship-radiated noise are analyzed, and the corresponding signal model is established. First, the line-spectrum, continuous-spectrum, and modulation components are introduced according to their physical generation mechanisms. Subsequently, a nonuniform modulation model is developed to simulate practical ship-radiated noise. Finally, a typical realization of the simulated signal is presented to illustrate its time-domain and frequency-domain characteristics.

Ship-radiated noise serves as the primary carrier of modulation information in DEMON analysis. Its statistical characteristics and spectral structures follow specific physical mechanisms and therefore cannot be sufficiently described by arbitrary stochastic processes. Consequently, an appropriate signal model is required for the analysis and simulation of DEMON spectra.

Ship-radiated noise originates from multiple mechanical structures and acoustic radiation mechanisms. According to previous studies, the dominant noise sources can generally be classified into mechanical noise, propeller noise, and hydrodynamic noise [29]. In the spectral domain, ship-radiated noise consists of discrete line spectra and broadband continuous spectra, with additional modulation components caused by periodic mechanical operations. Therefore, the signal model is established from these three aspects.

The line-spectrum component exhibits strong periodicity and can be regarded as the superposition of several fundamental frequency components and their harmonics. Therefore, the line-spectrum component can be expressed as

$$s_l(t) = \sum_{k=1}^{K} a_k \sin(2\pi f_k t + \varphi_k) \tag{1}$$

where a_k, f_k, and φ_k denote the amplitude, frequency, and random phase of the k-th spectral line, respectively. The random phase is assumed to be uniformly distributed over $[0, 2\pi]$. In general, the level of the line-spectrum component is approximately 10–25 dB higher than that of the continuous-spectrum component. The line spectra mainly originate from mechanical noise, propeller singing noise, and blade passing frequency components.

The continuous-spectrum component is mainly dominated by propeller cavitation noise, whereas the contributions of mechanical noise and hydrodynamic noise are relatively weak or masked by cavitation noise at high frequencies. According to the ideal cavitation spectrum, the continuous-spectrum component possesses a wide bandwidth, and its spectral peak is generally located between 100 and 1000 Hz [30].

The rotation of the propeller shaft and the periodic vibration of the blades introduce modulation effects into the radiated noise, resulting in periodic amplitude fluctuations in the time domain. The modulation spectrum contains characteristic components such as shaft frequency, blade passing frequency, and their harmonics, which provide valuable information for target classification and speed estimation.

Several studies model the modulation envelope of ship-radiated noise as a random pulse process with identical shapes and repetition periods but random amplitudes. An individual pulse is usually represented by a Gaussian function, as given in

$$\mu_\xi(t) = \frac{\xi}{2\pi} e^{-\frac{t^2}{2\sigma^2}} \tag{2}$$

where ξ and σ denote the amplitude of the pulse and the pulse-width parameter, respectively. The pulse-width parameter satisfies

$$\sigma = \frac{T}{BD} \tag{3}$$

where T, B and D denote the propeller shaft period, the number of propeller blades and a parameter controlling the pulse-width, respectively.

The interval between adjacent pulses depends on the modulation characteristics. For a propeller with B blades, the pulse spacing is determined by the blade-passing period. The pulse amplitudes are assumed to be mutually independent and follow a prescribed probability distribution. Consequently, the modulation component can be expressed as

$$s_m(t) = \sum_{i=0}^{\infty} \sum_{j=0}^{B-1} \mu_{\xi_{ij}} \left(t - iT - j\frac{T}{B} \right) \tag{4}$$

where $\mu_{\xi_{ij}}$ denotes the j-th pulse of the i-th group. Its amplitude is assumed to be uniformly distributed over $\left[\overline{\xi_j}/2, 3\overline{\xi_j}/2\right]$.

Based on the above assumptions, the time-domain model of ship-radiated noise can be represented by

$$n(t) = [1 + s_m(t)]s_c(t) + s_l(t) \tag{5}$$

where $s_m(t)$, $s_c(t)$ and $s_l(t)$ denote modulated spectral noise, continuous spectral noise and line spectral noise, respectively [31].

The model described by (5) corresponds to a uniformly modulated noise process. However, practical ship-radiated noise generally exhibits nonuniform modulation characteristics, where different frequency bands of the continuous-spectrum component possess different modulation depths. Therefore, a nonuniform modulation model is considered in this paper.

Since the modulation component mainly acts on the continuous-spectrum component, only the modulated continuous-spectrum term is analyzed [32]. According to Fourier theory, an arbitrary signal can be represented as the superposition of infinitely many complex exponential components. Consequently, the continuous-spectrum component can be expressed as

$$s_c(t) = \int_{-\infty}^{\infty} S_c(f) e^{j2\pi f t} df \tag{6}$$

For each frequency component, a frequency-dependent coefficient is introduced to characterize the modulation depth. The corresponding modulation result can then be expressed as

$$X(f,t)df = [1 + a(f)s_m(t)]S_c(f)e^{j2\pi f t}df \tag{7}$$

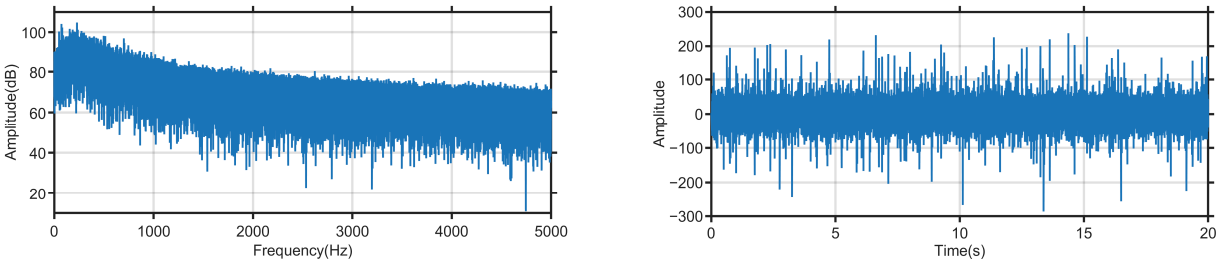
where $a(f)$ denotes the modulation-depth parameter.

By integrating over the entire frequency range, the nonuniformly modulated continuous-spectrum component can be obtained as

$$x(t) = s_c(t) + s_m(t)\mathcal{F}^{-1}[a(f)S_c(f)] \tag{8}$$

Finally, the complete nonuniform modulation model of ship-radiated noise is established by combining the modulated continuous-spectrum component with the line-spectrum component.

The time-domain waveform and frequency spectrum of the simulated ship-radiated noise are shown in Figure 1.



(**a**) (**b**)

Figure 1. Simulation results of ship-radiated noise: (**a**) signal frequency spectrum; (**b**) signal time-domain waveform.

3. Proposed Method

In this section, a noise-robust DEMON spectrum extraction framework integrating adaptive decomposition and sparse reconstruction is presented. The proposed framework aims to overcome the limitations of empirical frequency-band selection, background interference, and weak line-spectrum visibility in conventional DEMON spectrum extraction. First, EEMD combined with a normalized modulation intensity coefficient-based IMF selection strategy is employed to adaptively extract modulation-related components. Then, the diagonal slice of the third-order cumulant is introduced to obtain a noise-resistant signal representation by exploiting higher-order statistical characteristics. Finally, sparse Bayesian learning is utilized to enhance weak line spectra and obtain the desired DEMON spectrum estimate.

The proposed method consists of three sequential stages, namely adaptive component selection, noise-resistant signal representation construction, and sparse line-spectrum enhancement. The details of each stage are described in the following subsections.

3.1. Adaptive Signal Reconstruction

The basic idea of EEMD is to perform multiple independent noise-assisted decomposition processes on the original signal. By introducing Gaussian white noise with finite amplitude, the distribution of extrema becomes more uniform and the characteristic scales become more complete, thereby effectively alleviating envelope distortion and mode mixing [33]. Taking advantage of the zero-mean property of Gaussian white noise, the corresponding IMFs obtained from different decomposition trials are averaged to eliminate the influence of the added noise while preserving the intrinsic oscillatory modes of the original signal. For ship-radiated noise signals, modulation components related to shaft rotation and propeller motion may be distributed among different oscillatory modes. Therefore, EEMD is employed to obtain multiple intrinsic mode functions with different time-scale characteristics, providing a basis for subsequent signal reconstruction.

Specifically, K groups of independent Gaussian white noise are added to the original signal to generate a series of noise-assisted signals

$$x_i(t) = s(t) + n_i(t) \tag{9}$$

where $x_i(t)$, $s(t)$ and $n_i(t)$ denote the i-th noisy signal, the decomposed signal, and the i-th Gaussian noise, respectively.

Each noise-assisted signal is then decomposed by EMD to obtain a set of IMFs and a residual component

$$x_i(t) = \sum_{j=1}^{J} c_{i,j}(t) + r_i(t) \tag{10}$$

where J, $c_{i,j}(t)$ and $r_i(t)$ denote the number of IMFs, the j-th IMF component of the i-th noisy signal and the residual component, respectively of the i-th noisy signal.

Finally, the corresponding IMFs and residual terms obtained from different trials are averaged to obtain

$$c_j(t) = \frac{1}{K}\sum_{i=1}^{K} c_{i,j}(t) \tag{11}$$

Accordingly, the original signal can be represented as the sum of the averaged IMFs and the residual component.

After obtaining a series of IMFs, an IMF frequency-based screening strategy is first applied to remove components that cannot satisfy the modulation relationship of ship-radiated noise. According to the analysis of the inherent physical characteristics and modulation spectrum characteristics of ships in [34], the shaft frequency is generally distributed within 1–20 Hz, and the number of propeller blades is typically within the range of 3–7. Therefore, the maximum blade frequency can be estimated as 140 Hz.

To ensure that the retained IMF components can serve as carrier signals for blade-frequency modulation, their center frequencies should be higher than the blade-frequency range. Therefore, the center frequency of each IMF is calculated, and the components with center frequencies lower than a preset threshold are discarded. Considering the frequency estimation error and the variation in practical ship-radiated noise, a threshold of 200 Hz is adopted in the simulations. The robustness of this threshold selection is further discussed in the subsequent subsection.

The modulation intensity coefficient of the j-th IMF is defined as

$$R_j = \max\{|\mathcal{F}[|\mathcal{H}[c_j(t)]|]|\} \tag{12}$$

The modulation intensity coefficients are subsequently normalized using the maximum value among all IMF components. The normalized modulation intensity coefficient is defined as

$$\widetilde{R}_j = \frac{R_j}{\max\limits_{1\le j\le K}\{R_j\}} \tag{13}$$

where K denotes the total number of IMF components.

A larger normalized modulation intensity coefficient indicates that the corresponding IMF contains stronger modulation characteristics in its Hilbert envelope spectrum, whereas a smaller value indicates weaker modulation information. To automatically identify the IMFs with significant modulation components, the sum of the mean value and standard deviation of all normalized modulation intensity coefficients is employed as the adaptive selection threshold.

$$\lambda = \overline{\widetilde{R}} + \sqrt{\frac{1}{K-1}\sum_{j=1}^{K}\left(\widetilde{R}_j - \overline{\widetilde{R}}\right)^2} \tag{14}$$

The IMFs whose normalized modulation intensity coefficients exceed the threshold are then summed to obtain the reconstructed signal, which is used as the observation signal for subsequent processing. This reconstruction procedure suppresses IMFs with weak modulation characteristics and enhances the modulation-related components contained in the signal, thereby providing a more suitable representation for subsequent DEMON spectrum extraction. Through this procedure, IMFs with insufficient modulation information are discarded, while the components containing stronger modulation characteristics are retained for further analysis.

3.2. Gaussian Noise Suppression

After adaptive reconstruction, the retained signal components are further processed to extract modulation characteristics. Since DEMON analysis focuses on periodic fluctuations contained in the signal envelope, the Hilbert transform is applied to obtain the envelope signal of the reconstructed sequence.

For an arbitrary zero-mean stationary Gaussian process, the third-order cumulant is identically zero [35], which can be expressed as

$$c_3(\tau_1, \tau_2) = \mathbb{E}[x(t)x(t+\tau_1)x(t+\tau_2)] \equiv 0 \tag{15}$$

Therefore, the diagonal slice of the third-order cumulant, where $\tau = \tau_1 = \tau_2$, is also identically zero. Compared with higher-order cumulants, the third-order cumulant requires lower computational complexity while retaining the ability to suppress zero-mean Gaussian noise. Therefore, it provides an appropriate trade-off between noise robustness and computational efficiency for subsequent spectral estimation.

Accordingly, the diagonal slice of the third-order cumulant is adopted as the observation signal for subsequent sparse spectral estimation, where the influence of Gaussian noise is reduced while modulation-related components are preserved.

3.3. Spectral Enhancement

After Gaussian noise suppression, the modulation components of interest are expected to exhibit sparse characteristics in the frequency domain. This property is consistent with the spectral characteristics of DEMON signals, where shaft-related and blade-passing-

related components usually appear as discrete spectral peaks rather than continuous broadband components. Therefore, the diagonal slice of the third-order cumulant can be modeled as a sparse combination of frequency atoms, and sparse Bayesian learning is introduced to estimate the corresponding spectral coefficients for improving the representation of weak modulation components.

Based on the sparse spectral characteristics of DEMON signals, the observation sequence obtained from the diagonal slice of the third-order cumulant can be represented as a sparse combination of frequency atoms. Specifically, the observation model is expressed as

$$\boldsymbol{y} = \boldsymbol{\Phi}\boldsymbol{\omega} + \boldsymbol{\epsilon} \tag{16}$$

where $\boldsymbol{y}$ denotes the discrete observation sequence with length N, $\boldsymbol{\Phi}$ is the dictionary matrix consisting of M frequency atoms, $\boldsymbol{\omega}$ represents the sparse weight vector to be estimated, and $\boldsymbol{\epsilon}$ denotes the residual noise term. Each column of the dictionary matrix corresponds to a complex exponential basis associated with a frequency grid, which can be written as

$$\boldsymbol{\Phi} = [\Phi_1, \Phi_2, \ldots, \Phi_i, \ldots, \Phi_M] \tag{17}$$

$$\Phi_i = \left[1, e^{j2\pi f_i \frac{1}{N}}, \ldots, e^{j2\pi f_i \frac{N-1}{N}}\right]^T \tag{18}$$

The key idea of sparse Bayesian learning is to assign hierarchical priors to the weight vector, allowing the algorithm to automatically determine which coefficients should be retained and which should be driven to zero. Specifically, the weights are assumed to follow a complex Gaussian prior

$$p(\boldsymbol{\omega}\,|\boldsymbol{\alpha}\,) = \mathcal{CN}\left(\boldsymbol{\omega}\middle|0, \boldsymbol{A}^{-1}\right) \tag{19}$$

where $\boldsymbol{A} = \mathrm{diag}(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_M)$ is a diagonal matrix containing the precision hyperparameters, and α_i denotes the precision hyperparameter associated with the i-th coefficient.

When α_i tends to infinity, the variance of the corresponding Gaussian distribution approaches zero, and the associated coefficient is gradually suppressed to zero. To further promote sparsity, Gamma hyperpriors are imposed on the precision parameters and the noise variance

$$p(\boldsymbol{\alpha}) = \prod_{m=1}^{M} \Gamma(\alpha_i|a, b\,) \tag{20}$$

$$p(\beta) = \Gamma(\beta_i|c, d\,) \tag{21}$$

where $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_i, \ldots, \alpha_M]^T$ contains the weight precision hyperparameters, and β denotes the noise precision hyperparameter.

By applying the type-II maximum likelihood criterion, the hyperparameters are optimized by maximizing the marginal likelihood. Under the estimated hyperparameters, the posterior distribution of the weight vector is obtained as a Gaussian distribution, whose covariance matrix and mean vector are given by

$$\boldsymbol{\Sigma} = \left(\beta\boldsymbol{\Phi}^H\boldsymbol{\Phi} + \boldsymbol{A}\right)^{-1} \tag{22}$$

$$\boldsymbol{\mu} = \beta\boldsymbol{\Sigma}\boldsymbol{\Phi}^H\boldsymbol{y} \tag{23}$$

where $\boldsymbol{\Sigma}$ and $\boldsymbol{\mu}$ denote the posterior covariance matrix and posterior mean vector, respectively [36].

The effective number of parameters for each coefficient is defined as γ_i, which is given by

$$\gamma_i = 1 - \alpha_i \Sigma_{ii} \tag{24}$$

where Σ_{ii} denotes the i-th diagonal element of the posterior covariance matrix.

Accordingly, the update equations of the hyperparameters are obtained as

$$\alpha_i^{new} = \frac{\gamma_i}{|\mu_i|^2} = \frac{1 - \alpha_i \Sigma_{ii}}{|\mu_i|^2} \tag{25}$$

and

$$\beta^{new} = \frac{N - \sum_i \gamma_i}{\|\boldsymbol{y} - \boldsymbol{\Phi}\boldsymbol{\mu}\|^2} \tag{26}$$

The hyperparameters are iteratively updated until convergence. During the iteration, coefficients associated with excessively large precision parameters are removed to reduce the dimensionality of the dictionary and improve computational efficiency.

After convergence, the final spectral estimation strategy is selected according to whether the estimated frequency grid can be represented by the discrete Fourier transform (DFT) sampling frequencies. When all frequency points of interest in the constructed grid correspond to available DFT sampling frequencies, the signal is reconstructed using the estimated sparse coefficients, and the spectrum is obtained through Fourier analysis of the reconstructed signal. Otherwise, the posterior mean vector is directly adopted as the spectral estimate to preserve the high-resolution frequency information provided by the sparse reconstruction.

According to the modulation characteristics of ship-radiated noise described above, the frequency range of the SBL dictionary is determined according to the interested modulation components. Since the blade frequency is generally lower than 140 Hz, a frequency range of 0–200 Hz is adopted in the simulations, which covers the shaft frequency, blade-frequency components, and their dominant harmonic lines. This strategy avoids constructing unnecessary basis functions outside the frequency range of interest and thus reduces the computational burden. Furthermore, the frequency grid density of the dictionary can be adjusted according to practical requirements. A denser frequency grid can be employed to obtain higher spectral resolution, whereas a relatively sparse grid can be selected when computational efficiency is prioritized.

3.4. Algorithm Implementation

3.4.1. Overall Procedure of the Proposed Method

Based on the above description, the procedure of the proposed method is summarized in Algorithm 1.

By sequentially processing the received signal according to the above procedures, the desired DEMON spectrum estimate can be obtained. The overall processing framework, including adaptive decomposition, noise suppression, and sparse spectral reconstruction, is illustrated in Figure 2.

3.4.2. Computational Complexity Analysis

According to the complexity analysis of EMD in [37], the time complexity of one EMD can be regarded as $\mathcal{O}(N \log N)$ when the number of sifting iterations is fixed. Since EEMD performs EMD for K noise-assisted realizations, the complexity of the EEMD stage is $\mathcal{O}(KN \log N)$.

Algorithm 1. Pseudocode of the proposed algorithm

Input: Received signal $\boldsymbol{x}$, sampling frequency f_s, frequency vector f, and ensemble number K of EEMD
Initialization: Set $\boldsymbol{\alpha} = 1_M$, $\beta = \frac{0.01}{var(\boldsymbol{x})}$ and $r = 0$;
For i = 1, 2, . . ., K
Add Gaussian white noise to the original signal.
Perform EMD decomposition.
End For
Reconstruct the signal $\boldsymbol{x}_{re}$ using the selected IMF components
Extract the envelope signal $\widetilde{\boldsymbol{x}}_{re}$ of the reconstructed signal using Hilbert transform.
Calculate the diagonal slice of the third-order cumulant $\boldsymbol{c}_3(\tau)$ from the envelope signal.
Construct the dictionary matrix $\boldsymbol{\Phi}$ according to the frequency vector f.
While $\frac{\|\boldsymbol{\alpha}^{(r+1)} - \boldsymbol{\alpha}^r\|}{\|\boldsymbol{\alpha}^r\|} \geq 10^{-2}$
Calculate $\boldsymbol{\Sigma}$ according to Equation (21).
Calculate $\boldsymbol{\mu}$ according to Equation (22).
Update $\boldsymbol{\alpha}$ according to Equation (24), and remove the corresponding frequency components if α_i exceeds the predefined threshold.
Update β according to Equation (25).
Set $r = r + 1$.
End While
If all frequency points in f correspond to available DFT sampling frequencies
$\boldsymbol{y} = \mathcal{F}[\boldsymbol{\Phi}\boldsymbol{\mu}]$
Else
$\boldsymbol{y} = \boldsymbol{\mu}$
End If
Output: DEMON spectrum estimate $\boldsymbol{y}$

In practical implementation, the envelope signal is first obtained through Hilbert demodulation. The Hilbert transform can be efficiently performed using Fast Fourier Transform (FFT) operations, resulting in a computational complexity of $\mathcal{O}(N \log N)$. Subsequently, the diagonal slice of the third-order cumulant is calculated by evaluating the cross-correlation between the envelope signal and its squared envelope signal. The point-wise squaring operation requires $\mathcal{O}(N)$, while the cross-correlation can be efficiently computed using FFT operations with a complexity of $\mathcal{O}(N \log N)$. Therefore, the overall computational complexity of this stage is $\mathcal{O}(N \log N)$.

In each iteration of SBL, the main computational burden comes from the update of the posterior covariance matrix and posterior mean. For a frequency dictionary with M grid points and an observation sequence of length N, the matrix multiplication and inversion operations introduce computational complexities of $\mathcal{O}(NM^2)$ and $\mathcal{O}(M^3)$, respectively. Therefore, the overall computational complexity of the SBL stage is $\mathcal{O}(R(NM^2 + M^3))$, where R denotes the number of iterations.

Therefore, the overall computational complexity of the proposed method can be expressed as $\mathcal{O}(KN \log N + R(NM^2 + M^3))$, where K and R denote the number of EEMD realizations and SBL iterations, respectively, and M represents the number of frequency grid points.

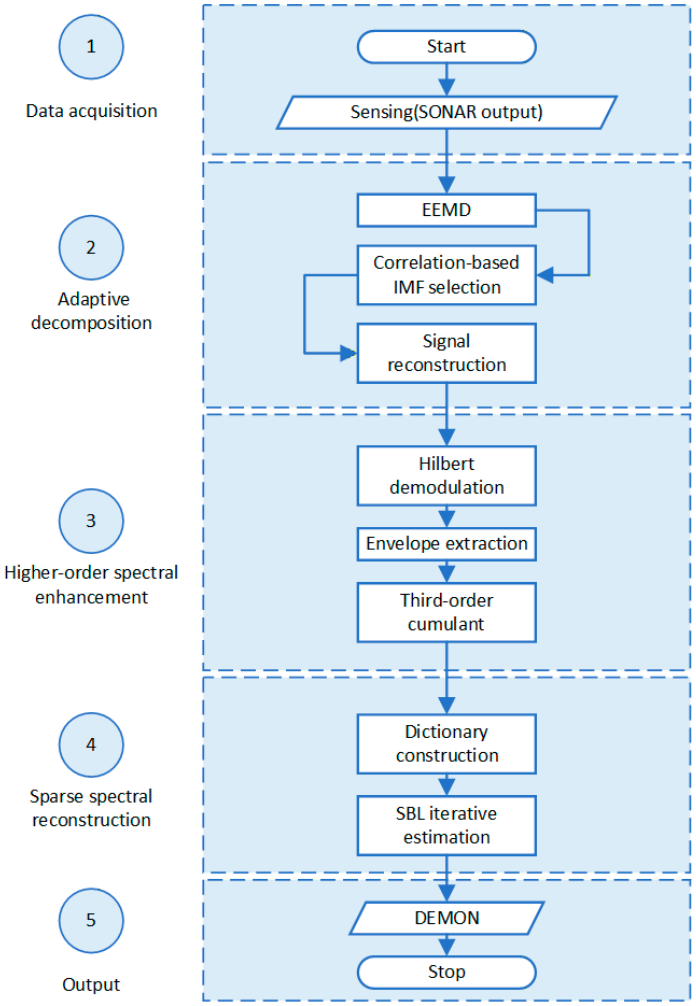


Figure 2. Flowchart of the proposed DEMON spectrum extraction method.

4. Simulation and Analysis

In this section, numerical simulations are conducted to evaluate the performance of the proposed method. First, the simulation data generation process and parameter settings are described based on the ship-radiated noise model established previously. Subsequently, the proposed method is compared with an existing DEMON spectrum enhancement approach, and an ablation analysis is conducted to investigate the contribution of different modules in the proposed framework. Finally, the robustness of the proposed method is evaluated under different noise statistical characteristics and selected parameter settings, and the convergence behavior of the SBL algorithm is discussed.

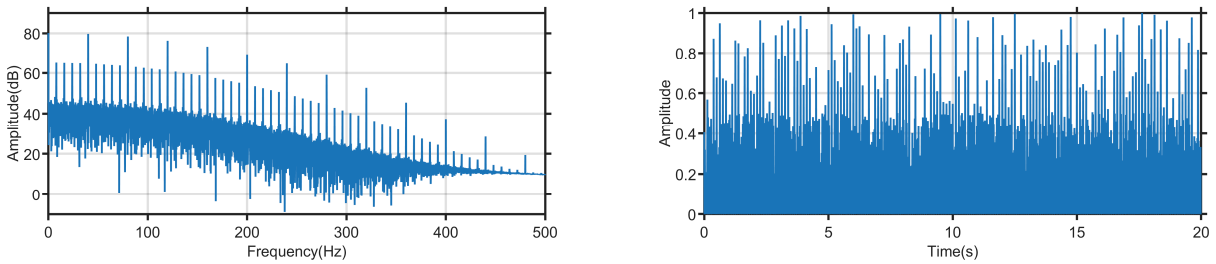
4.1. Simulation Data Generation and Evaluation Criteria

Considering that underwater acoustic noise may exhibit non-Gaussian statistical characteristics, the generalized Gaussian distribution (GGD) has been adopted to describe underwater acoustic noise with different statistical properties [38]. Compared with the Gaussian distribution, the GGD provides a more flexible statistical model through the adjustment of the shape parameter β. Therefore, in this simulation, the continuous-spectrum component of the received signal is generated from a GGD random sequence and subsequently filtered according to the prescribed spectral shape, where the shape parameter β is set to 1.8.

In the simulation, the sampling frequency and signal duration are set to 10 kHz and 20 s, respectively. The power frequency is set to 50 Hz, the shaft frequency is set to 8 Hz, and the number of propeller blades is set to 5. Six propeller singing spectral lines are randomly selected. Except for these randomly selected singing components, the line-spectrum frequencies are set as 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1100, and 1200 Hz. The peak frequency of the continuous-spectrum component is set to 200 Hz. The spectral level increases at a rate of 6 dB/oct below the peak frequency and decreases at the same rate above the peak frequency.

For the modulation model, the pulse width parameter D in (3) is set to 4, and the mean amplitudes of the pulses in each group are set to [1, 0.5, 0.5, 0.5, 0.5]. The modulation depth weighting factor is assigned its maximum value for frequency components below 1000 Hz and decreases with frequency at a rate of 80 dB/oct above 1000 Hz.

Based on the above simulation conditions, the time-domain waveform of the modulation function and its corresponding frequency spectrum are shown in Figure 3.



(**a**) (**b**)

Figure 3. Simulation results of the modulation signal: (**a**) signal frequency spectrum; (**b**) signal time-domain waveform.

To evaluate the performance of the proposed method under different modulation depth conditions, two simulation cases are considered, where the maximum modulation parameter is set to 3 and 6, respectively. The corresponding modulation depths are calculated as 0.75 and 0.6, respectively.

According to the evaluation criteria for ship-radiated noise DEMON spectra proposed in [39], the shaft-frequency harmonic line-spectrum signal-to-noise ratio, the clarity of the shaft frequency and its harmonics in the first harmonic cluster, and the blade-frequency clarity are employed as evaluation metrics. Among these metrics, the line-spectrum clarity (LSC) is used to quantitatively characterize the prominence of spectral lines and is defined as

$$\sigma_s = \frac{P_s}{\frac{1}{l}\sum_{i=0}^{l} \exp\left(-\frac{|f_N(i) - f_s|}{f_0}\right) \cdot P_N(i)} \tag{27}$$

where P_s and f_s denote the amplitude and frequency of the line spectrum under consideration, respectively, and f_0 denotes the shaft frequency. l represents the number of interference peaks located within the frequency interval $(f_s - 0.5 \times f_0, f_s + 0.5 \times f_0)$, while $P_N(i)$ and $f_N(i)$ denote the amplitude and frequency of the i-th interference peak, respectively.

In addition to the LSC, the shaft-frequency harmonic line signal-to-noise ratio (SHLSNR) is employed to evaluate the detectability of shaft-frequency harmonics. In this study, the SNR is calculated within the frequency range of the first shaft-frequency harmonic cluster for convenience of quantitative analysis, and its definition is given as

$$S_N = \frac{S}{\sum_{i=0}^{P} y(i) - S} \tag{28}$$

where S denotes the sum of the amplitudes of the shaft-frequency harmonic components within the considered frequency range.

4.2. Performance Comparison and Ablation Analysis

To validate the effectiveness of the proposed method, comparative experiments are conducted with the conventional EMD-based method and the DEMON spectrum extraction method based on improved EMD and $1\frac{1}{2}$ dimensional spectrum analysis ($1\frac{1}{2}$D) reported in [21]. The DEMON spectrum extraction results of the proposed method, the conventional EMD method, and the method in [21] are presented in Figure 4a–c, respectively.

Furthermore, ablation experiments are performed to investigate the contribution of each module in the proposed framework. Three simplified versions, including EEMD-$1\frac{1}{2}$D, EEMD-SBL, and TCDS-SBL, are compared with the complete method, and the corresponding results are shown in Figure 4d–f, respectively. Here, TCDS denotes the diagonal slice of the third-order cumulant. The SHLSNR and LSC of each harmonic component of the six methods are quantitatively evaluated and summarized in Table 1.

As shown in Figure 4a,c, the proposed method achieves better DEMON spectrum extraction performance than the method reported in [21]. Specifically, the proposed method exhibits a lower background noise level and stronger harmonic LSC. According to the quantitative results in Table 1, the SHLSNR of the proposed method is improved by approximately 87 times compared with that of the method in [21], while the clarity of the extracted harmonic components is increased by 3.1–12.4 times.

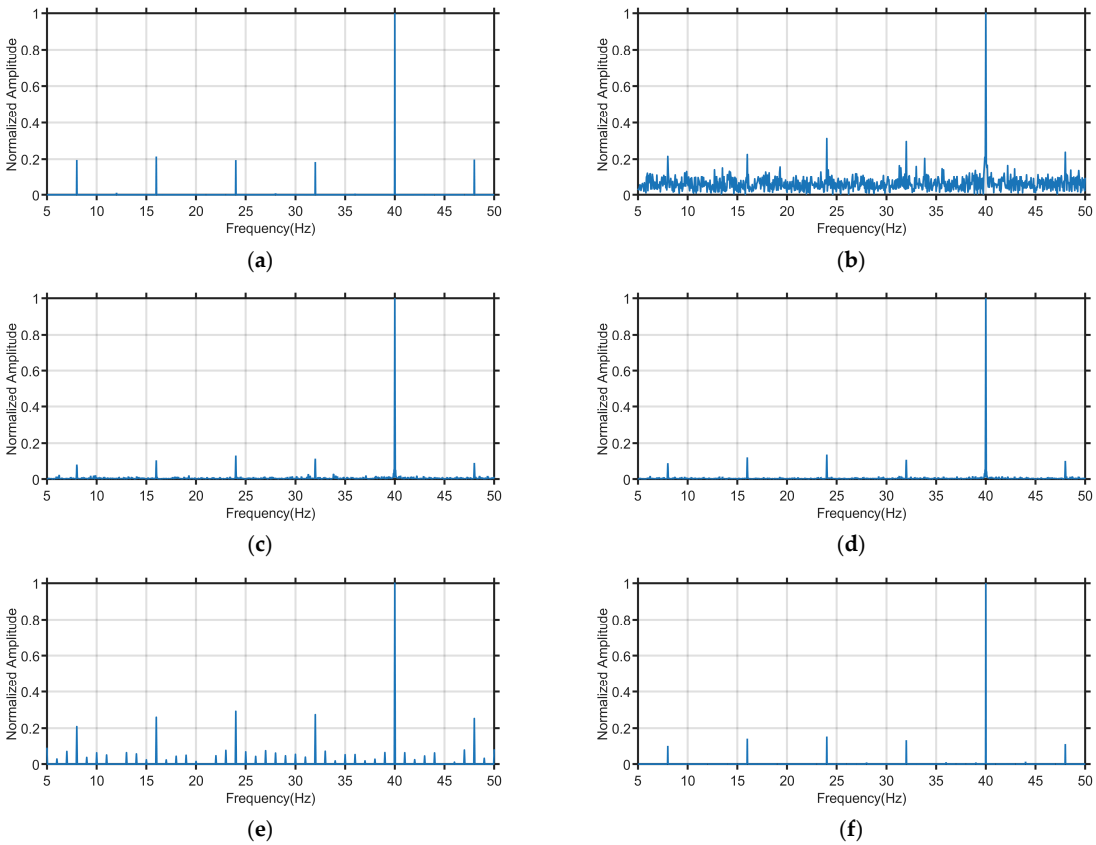


Figure 4. Comparison of DEMON spectra under a maximum modulation depth of 0.75. (**a**) Proposed method. (**b**) EMD-based method. (**c**) EMD-$1\frac{1}{2}$D method. (**d**) EEMD-$1\frac{1}{2}$D method. (**e**) EEMD-SBL method. (**f**) TCDS-SBL method.

As summarized in Table 1, for the same received signal, the overall performance ranking of the compared methods is the proposed method, TCDS-SBL, EEMD-SBL, EEMD-

$1\frac{1}{2}$D, the method in [21], and the conventional EMD-based method. The ablation results in Figure 4d–f further indicate the effectiveness of each module in the proposed framework. The EEMD-based IMF selection strategy effectively preserves IMFs with stronger modulation-related characteristics, providing a more suitable signal representation for subsequent processing. The third-order cumulant diagonal slice exploits the noise-resistant property of higher-order statistics to suppress Gaussian noise components and enhance modulation-related features, while the SBL algorithm further improves the spectral concentration by exploiting the sparsity of line-spectrum components. Therefore, removing any individual module results in degraded DEMON spectrum extraction performance.

Table 1. Quantitative comparison of DEMON spectrum extraction results under a maximum modulation depth of 0.75.

Method	SHLSNR	H1 * LSC	H2 LSC	H3 LSC	H4 LSC	H5 LSC
Proposed method	19.6312	301.2215	302.4197	299.7342	291.5595	306.4420
EMD-based method	0.0502	5.0045	4.8737	6.9208	6.1291	17.0493
EMD-$1\frac{1}{2}$D method	0.2256	24.1997	30.5874	38.6552	28.8405	98.3116
EEMD-$1\frac{1}{2}$D method	0.2767	30.8284	43.4884	45.4126	33.3304	105.5218
EEMD-SBL method	1.4773	72.8374	91.0666	81.5237	79.4764	128.1108
TCDS-SBL method	15.2264	294.6255	299.9203	290.8618	279.8837	300.9389

* H represents the harmonic order.

Subsequently, the case with a maximum modulation depth of 0.6 is investigated, and the corresponding comparison results are shown in Figure 5. The quantitative evaluation results of the six methods, including the SHLSNR and LSC values of individual harmonic components, are summarized in Table 2.

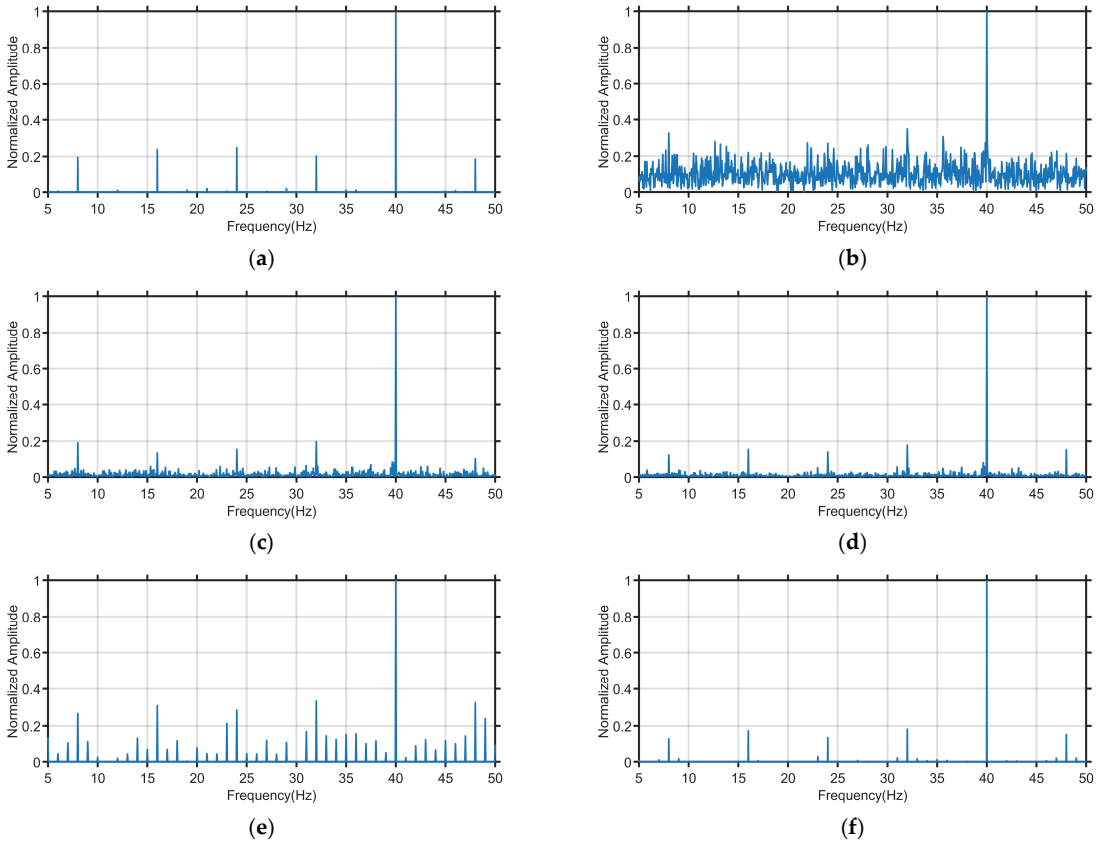


Figure 5. Comparison of DEMON spectra under a maximum modulation depth of 0.6. (**a**) Proposed method. (**b**) EMD-based method. (**c**) EMD-D method. (**d**) EEMD-D method. (**e**) EEMD-SBL method. (**f**) TCDS-SBL method.

As shown in Figure 5, with the decrease in modulation depth, the detection performance of all algorithms degrades to varying degrees. In particular, for the conventional EMD method, line spectra other than the blade-frequency component are gradually submerged in the background noise. In contrast, the proposed method maintains effective spectral line extraction capability under low modulation-depth conditions, with the LSC values of all harmonic components remaining above 250. The performance differences among the algorithms are consistent with the analysis in the previous section. Compared with the method in [21], the proposed method improves the LSC values of individual harmonic components by a factor of 5–20 and achieves an SHLSNR approximately 88 times higher. These results demonstrate that the proposed method maintains strong robustness even under low modulation-depth conditions. Furthermore, the ablation results indicate that each module in the proposed framework plays an important role in improving DEMON spectrum extraction performance.

Table 2. Quantitative comparison of DEMON spectrum extraction results under a maximum modulation depth of 0.6.

Method	SHLSNR	H1 LSC	H2 LSC	H3 LSC	H4 LSC	H5 LSC
Proposed method	8.4593	285.6408	282.5634	271.3759	254.9955	304.1823
EMD-based method	0.0305	4.6638	2.6664	3.4818	4.2105	11.3394
EMD-1$\frac{1}{2}$D method	0.0959	21.3292	13.6650	17.0690	18.4172	60.5199
EEMD-1$\frac{1}{2}$D method	0.1210	18.2959	21.4317	21.0665	21.5959	68.6916
EEMD-SBL method	0.8300	69.0208	69.5141	59.7955	53.02145	106.2347
TCDS-SBL method	6.2615	239.8544	291.0420	244.3796	229.4837	298.8539

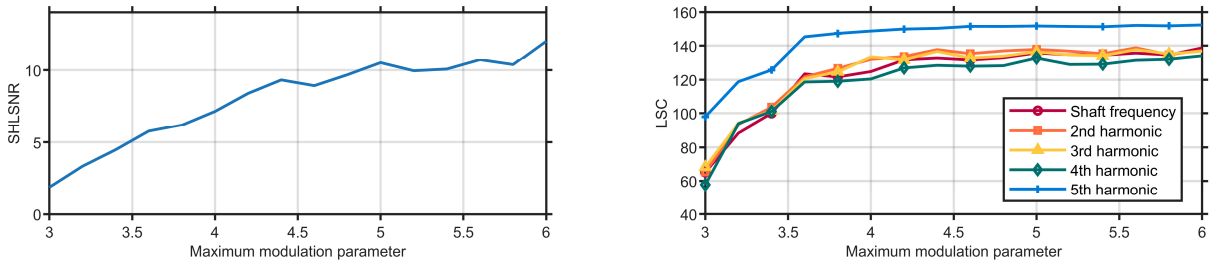
4.3. Robustness Analysis

To further investigate the robustness of the proposed method, additional experiments are conducted under different signal conditions and algorithm-related parameters. Since the SBL-based sparse reconstruction involves iterative optimization and relatively high computational cost, the signal duration is set to 10 s in the following experiments to improve computational efficiency. The selected duration still preserves the main characteristic components of ship-radiated noise, including shaft-frequency harmonics and blade-frequency components, enabling effective evaluation of the proposed method under different conditions.

4.3.1. Influence of Modulation Depth

The modulation depth is an important factor affecting the visibility of characteristic components in DEMON spectra. To evaluate the capability of the proposed method in extracting weak modulation features, the maximum value of the modulation parameter $a(f)$ is varied from 3 to 6, while other simulation parameters remain unchanged. For each modulation condition, 50 Monte Carlo trials are performed, and the averaged results are used for quantitative evaluation.

The LSC and SHLSNR values of the proposed method under different modulation conditions are shown in Figure 6. It can be observed that both evaluation metrics increase with the increase in the modulation parameter $a(f)$. Since a larger value of $a(f)$ corresponds to a higher modulation depth, the characteristic components in the DEMON spectrum become more prominent, resulting in improved LSC and higher SHLSNR. Even under the condition with a relatively low modulation depth, the LSC values of all harmonic components remain above 50, indicating that the characteristic spectral lines can still be effectively identified. These results demonstrate the robustness of the proposed method in extracting characteristic spectral components under different modulation conditions.



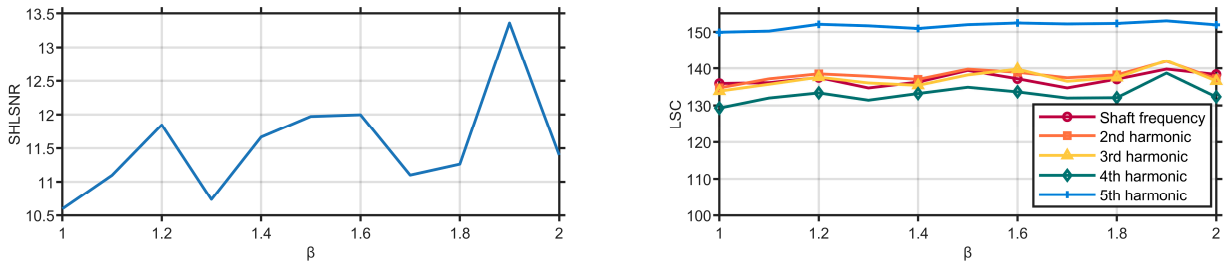
(**a**) (**b**)

Figure 6. Variation in SHLSNR and LSC with modulation parameter. (**a**) SHLSNR. (**b**) LSC.

4.3.2. Influence of GGD Shape Parameter

The statistical characteristics of the continuous-spectrum component are controlled by the shape parameter β of the GGD. In this experiment, β is varied from 1 to 2, where β = 2 corresponds to the Gaussian distribution, while smaller values of β indicate stronger non-Gaussian characteristics. The LSC and SHLSNR values are calculated under different β values to evaluate the robustness of the proposed method against different noise statistical characteristics. For each modulation condition, 50 Monte Carlo trials are performed, and the averaged results are used for quantitative evaluation.

The variation in the evaluation metrics with respect to β is shown in Figure 7. It can be observed that the SHLSNR exhibits a fluctuating increasing trend as β approaches 2, while the overall variation remains limited. Meanwhile, the LSC values of different harmonic components also show only limited variations over the investigated range of β. These results indicate that the statistical characteristics of the continuous-spectrum component have a limited influence on the DEMON spectrum extraction performance of the proposed method within the investigated range of β.



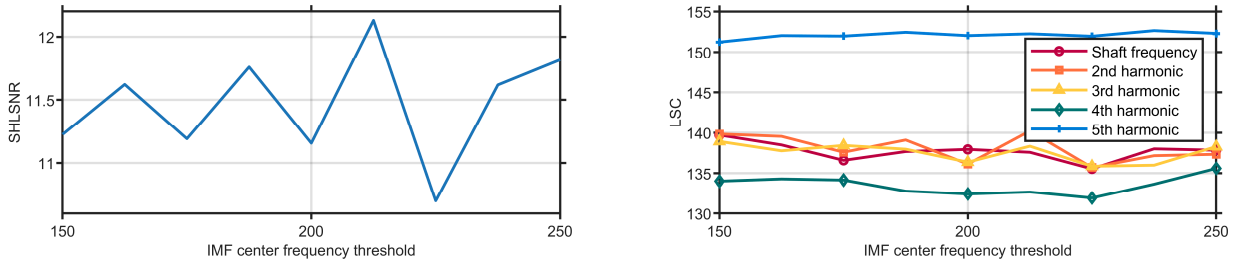
(**a**) (**b**)

Figure 7. Variation in SHLSNR and LSC with β. (**a**) SHLSNR. (**b**) LSC.

4.3.3. Influence of IMF Center Frequency Threshold

The IMF center frequency threshold directly affects the selection of carrier components for subsequent DEMON spectrum extraction. Therefore, the influence of this threshold is investigated to evaluate the robustness of the proposed method against variations in IMF selection. In the simulations, the threshold is varied from 150 Hz to 250 Hz around the preset value of 200 Hz, while other simulation parameters remain unchanged. For each threshold setting, 50 Monte Carlo trials are conducted, and the averaged LSC and SHLSNR values are used for quantitative evaluation.

The variation in LSC and SHLSNR with the IMF center frequency threshold is shown in Figure 8. It can be observed that both metrics exhibit only slight variations as the threshold changes. In particular, the variation in SHLSNR remains within 1.5 over the investigated range, indicating that the performance of the proposed method is insensitive to moderate changes in the IMF center frequency threshold. Therefore, selecting 200 Hz as the threshold provides a stable choice for practical DEMON spectrum extraction.



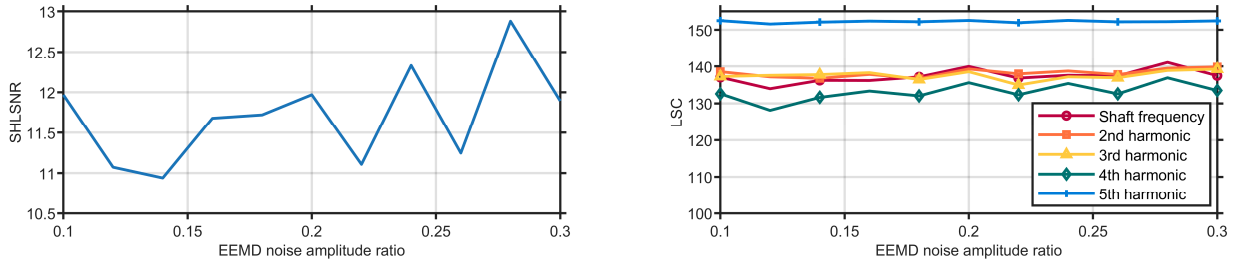
(**a**) (**b**)

Figure 8. Variation in SHLSNR and LSC with IMF center frequency threshold. (**a**) SHLSNR. (**b**) LSC.

4.3.4. Influence of EEMD Noise Amplitude

The added noise amplitude is an important parameter in the EEMD process, as it affects the decomposition results of the original signal. Therefore, the influence of the EEMD noise amplitude is investigated to evaluate the robustness of the proposed method with respect to this parameter. In the simulations, the amplitude of the added white noise is varied from 0.1 to 0.3 times the standard deviation of the original signal, while other simulation parameters remain unchanged. For each noise amplitude setting, 50 Monte Carlo trials are conducted, and the averaged LSC and SHLSNR values are used for evaluation.

The variation in LSC and SHLSNR with the normalized EEMD noise amplitude is shown in Figure 9. It can be observed that the DEMON spectrum extraction performance of the proposed method remains relatively stable as the noise amplitude ratio varies within the investigated range. Specifically, the variation in SHLSNR remains within 2, while the maximum variation in LSC is approximately 5. These results indicate that the proposed method is not significantly affected by the variation in the EEMD noise amplitude within the range of 0.1–0.3, and selecting 0.2 as the noise amplitude ratio provides a stable choice for practical applications.



(**a**) (**b**)

Figure 9. Variation in SHLSNR and LSC with EEMD noise amplitude ratio. (**a**) SHLSNR. (**b**) LSC.

4.3.5. Convergence Analysis of SBL

Since the proposed method employs SBL for sparse spectral reconstruction, the convergence behavior of the iterative optimization process is further investigated. The relative variation in hyperparameters is adopted as the convergence criterion, which is defined to evaluate the change in hyperparameters between successive iterations. The convergence behavior is analyzed based on the iterative results of the SBL algorithm.

The convergence curve of the SBL algorithm is shown in Figure 10. It can be observed that the relative variation in hyperparameters is relatively large during the first two iterations, indicating a rapid adjustment of the hyperparameters in the initial optimization stage. Subsequently, the relative variation quickly decreases and remains around a small stable value, demonstrating the convergence and stability of the iterative optimization process.

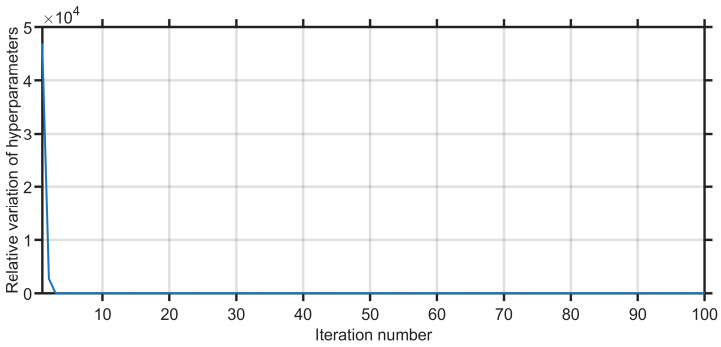


Figure 10. Convergence behavior of the proposed SBL algorithm.

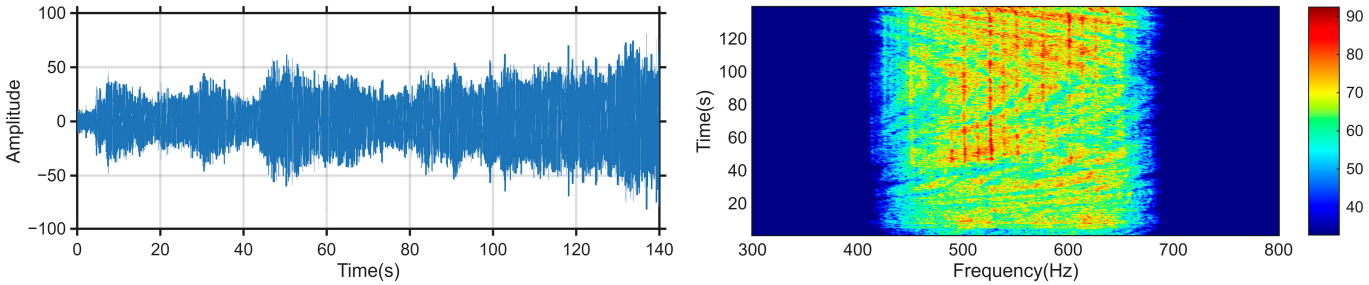
5. Experimental Results

In this section, measured ship-radiated noise data collected during a lake experiment are processed to further verify the effectiveness of the proposed method. First, the characteristics of the measured signal are illustrated through its time-domain waveform and LOFAR spectrum. Subsequently, the DEMON spectra obtained by the EMD-based method and the proposed method are compared to demonstrate the capability of the proposed method in enhancing weak modulation features under practical conditions.

5.1. Experimental Data

Measured ship-radiated noise data were collected during a lake experiment conducted in the Danjiangkou Reservoir. The sampling frequency was set to 13,333 Hz, and the target was a motorboat moving around the hydrophone in a circular trajectory. The time-domain waveform and LOFAR spectrum of the received acoustic pressure signal are presented in Figure 11.

To further investigate the statistical characteristics of the received data, the Lilliefors test was employed to examine its Gaussianity. The obtained p-value is 0.001, which is lower than the significance level of 0.05. Therefore, the hypothesis that the received data follow a Gaussian distribution is rejected, indicating that the received data exhibit non-Gaussian statistical characteristics.



(**a**) (**b**)

Figure 11. Time-domain waveform and LOFAR spectrum of the measured ship-radiated noise: (**a**) Time-domain waveform. (**b**) LOFAR spectrum.

5.2. DEMON Spectrum Comparison

The measured data are processed using the method proposed in [21] and the proposed method, respectively. The corresponding DEMON spectra are shown in Figure 12, and the quantitative evaluation results in terms of LSC and SHLSNR are listed in Table 3.

It can be observed from the measured data that both methods can identify the shaft frequency and its first seven harmonic components. However, the method in [21] exhibits a relatively high normalized background level. In particular, the fifth and seventh harmonics

of the shaft frequency are almost submerged by the background components, and their LSC values are both lower than 20, making these harmonic components difficult to distinguish. In contrast, the proposed method effectively suppresses most background interference components, resulting in a clearer separation between the harmonic components and the surrounding background. The LSC values of all harmonic components exceed 200, and the SHLSNR of the proposed method is approximately eight times that of the EMD-1$\frac{1}{2}$D method. These results demonstrate that the proposed method can effectively achieve adaptive frequency-band selection, noise suppression, and line-spectrum enhancement for measured ship-radiated noise signals.

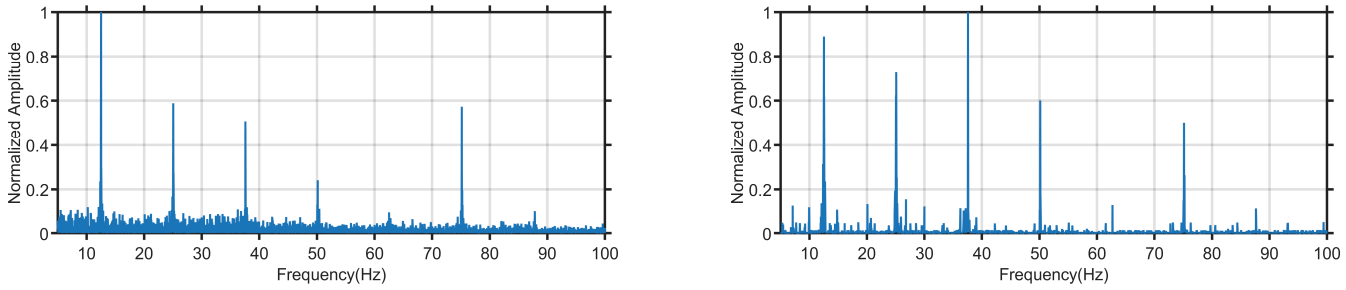


Figure 12. DEMON spectra obtained from measured data: (**a**) EMD-1$\frac{1}{2}$D method. (**b**) Proposed method.

Table 3. Quantitative comparison of LSC and SHLSNR between the proposed method and the EMD-1$\frac{1}{2}$D method.

Method	SHLSNR	H1 LSC	H2 LSC	H3 LSC	H4 LSC	H5 LSC	H6 LSC	H7 LSC
Proposed method	0.0850	242.2443	304.3448	350.1602	439.1095	305.9355	262.9120	206.8716
EMD-1$\frac{1}{2}$D method	0.0104	63.6990	44.7627	41.2896	28.5405	13.4041	69.0486	16.2219

6. Conclusions

In this paper, a noise-robust DEMON spectrum extraction method for ship-radiated noise based on adaptive decomposition and sparse reconstruction is proposed to improve the quality of DEMON spectra under strong background interference conditions. EEMD and an IMF selection strategy based on the normalized modulation intensity coefficient are employed to achieve adaptive frequency-band selection. The diagonal slice of the third-order cumulant is introduced to reduce the influence of Gaussian background components, and sparse Bayesian learning is utilized to enhance weak line spectra and improve the quality of the extracted DEMON spectrum. Simulations and measured-data processing are conducted to evaluate the performance of the proposed method, and the main conclusions are summarized as follows.

The simulation results demonstrate that the proposed method maintains satisfactory DEMON spectrum extraction performance under different modulation conditions and noise statistical characteristics. Compared with the method in [21], the proposed method provides a cleaner spectral background and clearer spectral lines. In particular, the SHLSNR achieved by the proposed method is improved by more than 80 times compared with the method in [21]. Furthermore, the ablation experiments demonstrate the contribution of the EEMD, third-order cumulant diagonal slice, and SBL modules, which further supports the necessity of each component and the rationality of the proposed processing framework.

Compared with the method in [21], the proposed method can effectively reduce the spectral background level and enhance the distinguishability of shaft-frequency harmonic components. The measured results show that although both methods can observe the first seven shaft-frequency harmonic components, the fifth and seventh harmonics obtained by the method in [21] are significantly affected by background interference, with LSC values lower than 20. In contrast, all harmonic components extracted by the proposed method achieve LSC values exceeding 200. Meanwhile, the SHLSNR achieved by the proposed method is approximately eight times that of the method in [21]. These results demonstrate that the proposed method can effectively realize adaptive frequency-band selection, background interference suppression, and line-spectrum enhancement, indicating its potential for DEMON spectrum extraction of ship-radiated noise under practical background interference conditions.

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Data Availability Statement: The data presented in this study are not publicly available due to privacy and confidentiality restrictions. The data presented in this study are available on request from the corresponding author.

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